
Unsupervised particle separation in a LArTPC via interpretable latent representations

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Abstract

Liquid argon time projection chambers (LArTPCs) are widely used in neutrino physics due to their ability to image charged-particle trajectories at high resolution. In LArTPC data, particles can be distinguished by their characteristic energy deposition profiles (calorimetry) and spatial patterns (topology). We ask whether an unsupervised model can discover and structure these differences directly from real detector data. We train a convolutional β -VAE to reconstruct 9,419 events from the LArIAT experiment and find that both calorimetry and topology are encoded in its eight-dimensional latent space. We then apply a Gaussian mixture model to the latent space to separate events into clusters corresponding to different particle species. Comparing these cluster assignments against particle-ID labels derived from auxiliary-detector information, we achieve a beamline-tag purity above 75% across all particle species. This first application of β -VAEs to real LArTPC data demonstrates that physically meaningful particle representations can be learned directly from detector data without particle labels in the loss function.

1 Introduction

Liquid argon time projection chambers (LArTPCs) are detectors used in state-of-the-art neutrino experiments [1, 12]. A charged particle travelling through the liquid argon leaves a trail of ionisation electrons which are then transported to a plane of wires using an electric field applied across the chamber. Each wire plane records a two-dimensional projection in wire-time coordinates, giving one image per plane for each event. Under this framework, particle identification can be thought of as a visual problem with two main cues: the shape of the trail (topology) and how much energy the particle deposits along the way (calorimetry). Broadly speaking, a stopping proton typically exhibits increasing energy deposition near its endpoint, a kaon decays and produces a secondary track, and a through-going minimum-ionising muon or pion (MIP) is usually straight with low energy deposition. Fig. 1 shows examples of these patterns.

In LArTPCs, machine learning is often used for segmentation, particle identification, and interaction classification [2–4, 13]. The vast majority of methods are supervised and obtain labels from detector simulation. Recent self-supervised approaches remove the need for labels, but still train on simulated LArTPC data [6, 15, 16], while generative models have primarily been applied to LArTPC images for event generation [9, 11]. Performance on real data is therefore still tied to how well the simulation reproduces the detector response [8]. A physically grounded representation learned directly from real images can sidestep this dependence. In this work, we test whether this is possible by training a variational autoencoder (VAE) [7, 10] to reconstruct real data from the Liquid Argon In A Testbeam (LArIAT) [5] experiment and perform an unsupervised particle-separation task. We test our results against particle identities determined by LArIAT’s auxiliary detectors upstream of the LArTPC. These particle identities are used to select samples that we label as proton, kaon, or MIP candidates, but neither labels nor other engineered features enter the loss function.

2 Data and model

We use positively charged LArIAT Run II data. Following cleaning cuts to remove noise and identify the primary particle, the sample contains 10,466 proton, 8,227 kaon, and 8,964 MIP candidates, as tagged by LArIAT’s auxiliary detectors. The auxiliary detectors provide a reconstructed beamline mass; the kaon tag selects events with masses between 350 and 650 MeV. Finite mass resolution allows some protons into this window, giving a kaon purity of $\sim 61\%$, while the proton and MIP tags are over 90% accurate (see Appendix C).

Each event has two wire-plane views, collection and induction, which are used as the two input channels. For image creation, we keep only the final 50 wires of the selected track. This window contains a stopping particle’s high-charge endpoint or a kaon’s secondary activity, while removing differences in full-track length. This choice retains key discriminating features at the end of the track while preventing the model from relying predominantly on track length for separation. Each view is placed on a 1502×51 zero-filled canvas and downsampled to 48×48 using bilinear interpolation. Charge values in analogue-to-digital converter (ADC) counts are transformed as $x \mapsto \log(1 + x)$ to reduce the dynamic range of the data. The VAE receives only the image, with no reconstructed track length, calorimetry, LArTPC detector calibration, or auxiliary-detector tags.

The convolutional β -VAE has four encoder blocks, an eight-dimensional Gaussian latent space, and a mirrored decoder. The loss combines a weighted mean squared reconstruction error with a Kullback–Leibler divergence to a unit Gaussian prior, weighted by $\beta = 0.5$ [7]. Active pixels receive ten times the weight of background pixels so that background does not dominate the reconstruction loss. The loss function is label-free; beamline labels are used only to construct a balanced training sample of 9,419 events. The remaining 18,238 events are held out from VAE training for validation. Appendix A reports checks on the data split and model capacity.

We use two simple observables to interpret the latent vectors. Mean ADC is the average recorded charge and serves as a calorimetry proxy. Solidity measures the fraction of a track’s convex hull occupied by active pixels and serves as a topology proxy: gaps, kinks, and secondary tracks lower the value. Within each tagged sample, we split validation events at the median of each proxy and fit a logistic regression probe to the latent vectors using five-fold cross-validation. The out-of-fold area under the receiver operating characteristic curve (AUC) measures how well each proxy can be decoded from the latent representation alone. We then fit full-covariance Gaussian mixture models (GMMs) to all 27,657 latent vectors. Labels, proxies, and the beamline mass estimate do not enter the GMM fit and are used only for evaluation and interpretation.

3 Results

3.1 Compression organises particle-dependent structure

We first inspect the representation before fitting a probe or clustering model. The top row of Fig. 1 shows representative detector images for the three particle species. The bottom row shows a t-SNE projection [14] of all latent vectors, with each species highlighted in turn. MIPs are concentrated along one side of the projection, kaons towards the centre, and protons across the remaining region, with substantial overlap between species. The t-SNE is used only to visualise this structure; all clustering is performed in the native eight-dimensional latent space.

We next test what this organisation represents. Figure 2 colours the same t-SNE projection by the two observables defined in Section 2. Mean charge is highest in one region, while solidity varies across the projection in a different pattern. This suggests that both energy deposition and track shape contribute to the organisation of the latent space. Neither observable was used to train the VAE.

We test this directly in the native eight-dimensional latent space (Fig. 3). Mean ADC is most strongly encoded for protons, with an AUC of 0.952, compared with 0.857 for kaons and 0.823 for MIPs. Solidity is most strongly encoded for kaons, with an AUC of 0.908, compared with 0.804 for protons and 0.788 for MIPs. Because each probe separates events above and below the median within a single tagged sample, these results cannot be explained simply by recovery of the auxiliary-detector labels. The species dependence is also physically consistent: protons exhibit the strongest charge variation, while kaon decays and interactions produce the largest variation in track shape. The latent representation therefore retains information about both calorimetry and topology.

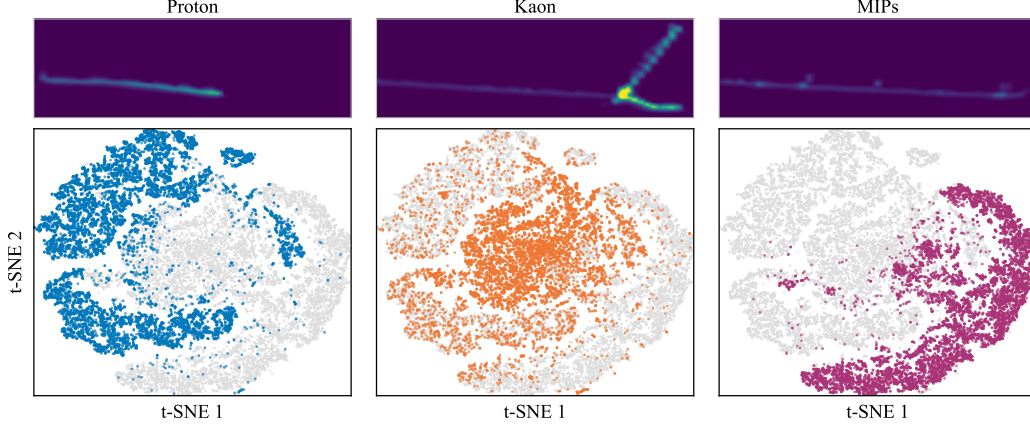


Figure 1: **Particle-dependent structure in the latent space.** Top: Collection-plane examples of a proton, kaon and a MIP. Bottom: the same t-SNE with each beamline label highlighted in turn.

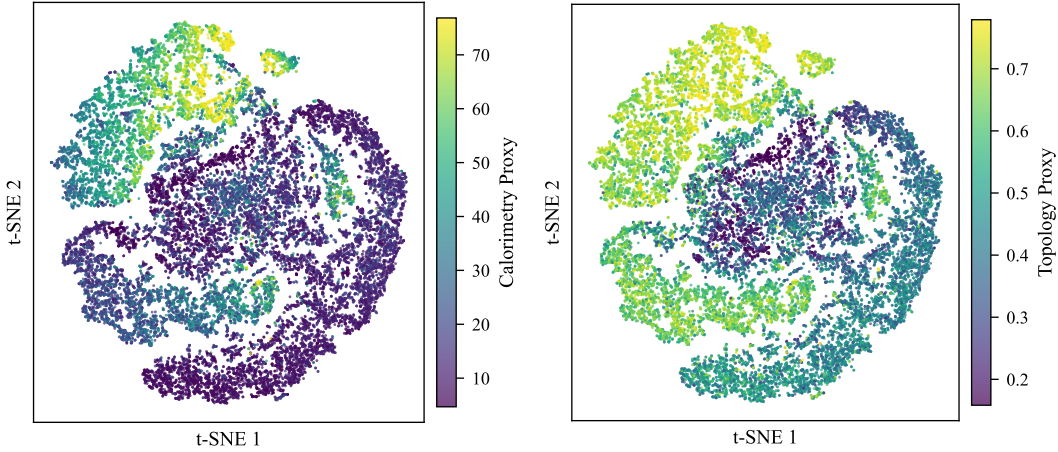


Figure 2: The shared t-SNE projection coloured by the calorimetry (left) and topology (right) proxy variables. Event positions are identical between panels.

3.2 Unsupervised clustering retrieves particle separation

We test whether the GMM can recover particle populations without seeing their tags. Agreement with the beamline labels from the auxiliary detectors plateaus across a scan over $k = 3\text{--}50$ clusters (Appendix B). We use $k = 37$ as a fine partition, not a unique physical decomposition, and label its clusters by species only after fitting. At $k = 37$, 22 clusters are at least 80% pure and contain 63.6% of all events. Five contain at least 90% kaon-tagged events and together contain 2,981 kaon candidates, or 36% of that sample, at 93.7% combined purity. Grouping clusters by their majority species and scoring each group against the beamline tags gives a purity of 86.6% for protons, 81.9% for MIPs, and 75.3% for kaons.

3.3 The representation isolates beam contamination

The beamline selection labels 8,227 candidates as kaons, but finite mass resolution allows proton contamination into the same mass window, so this label is not an event-level truth label. Since the kaon and proton masses are 493.7 and 938.3 MeV, respectively, proton contaminants reconstructed within the 350–650 MeV kaon selection window should preferentially populate its upper part. If the learned representation captures this contamination, kaon-tagged events with proton-like charge deposition and track shape should be encoded near the proton population and assigned to proton-majority GMM clusters. The beamline mass, which is not used by the VAE loss or GMM fit, therefore

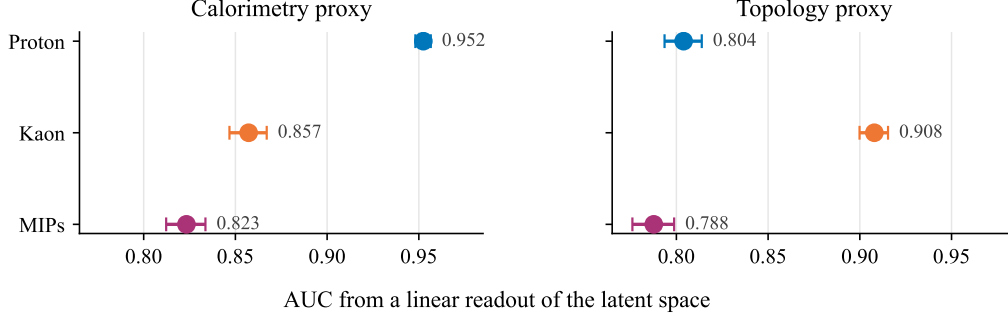


Figure 3: **Calorimetry and topology are linearly decodable from the latent representation.** Out-of-fold AUCs use species-specific median splits. Bars are 95% confidence intervals from resampling.

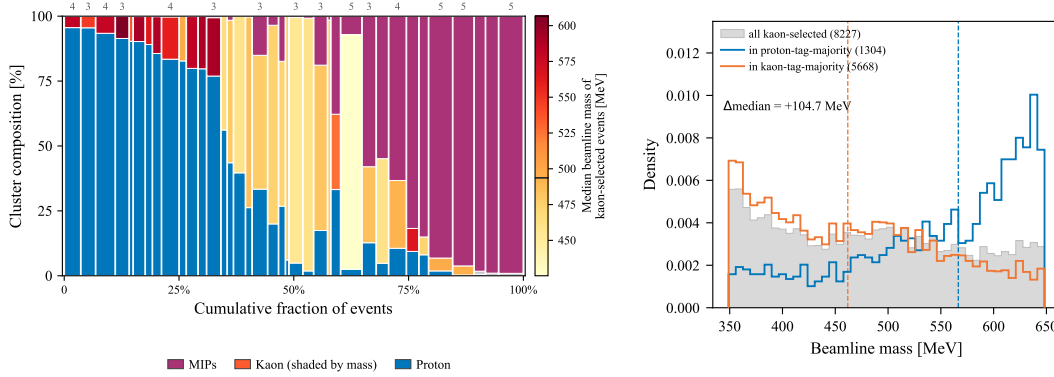


Figure 4: **Unsupervised clusters retrieve species populations and isolate contamination.** Left: $k = 37$ cluster composition, with width proportional to population and kaon segments shaded by median beamline mass. Right: beamline mass in the kaon-tagged selection split by cluster assignment.

provides an independent test of whether the latent representation separates these physically distinct populations. Appendix C details the beamline contamination.

Every event in this test has the same selection-derived kaon label. We examine these events within the same $k = 37$ latent-space clustering. Events assigned to proton-majority clusters have median beamline mass 566.5 MeV, compared with 461.9 MeV for those assigned to kaon-majority clusters, a shift of +104.7 MeV in the expected direction (Fig. 4). Across 28 clusters containing at least 50 kaon-tagged events, the median mass rises with the proton-tagged fraction (Spearman $\rho = 0.64$). No such correlation is found in the purer proton and MIP samples.

At $k = 37$, the proton-majority group contains 15.9% of the kaon-tagged sample, compared with an approximate 25.1% proton contamination scale estimated from the beamline distribution before the final data selection (Appendix C). The latent representation therefore isolates a physically consistent subset of the contamination without using the beamline label during clustering.

4 Discussion

To our knowledge, this is the first application of a variational autoencoder to real LArTPC data. With appropriate data preparation, the latent representation retains the two main features used to distinguish particle species in LArTPC images: energy deposition and track topology. Neither is specified in the loss function, yet both are recoverable from the latent vectors and support unsupervised separation into particle-enriched populations. This shows that useful particle representations can be learned directly from real detector images without defining these features by hand or using particle labels in the loss function.

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A Validation and robustness checks

A.1 Training and validation distributions

The balanced training set contains 3,139 protons, 3,140 kaons, and 3,140 MIPs. The validation set contains 7,327, 5,087, and 5,824 events, respectively. Before comparing pooled latent samples, the validation species mixture was matched to the training mixture. A classifier two-sample test obtained an AUC of 0.5026 ± 0.0029 for train versus validation codes, and an energy-distance permutation test gave $p = 0.64$. No latent coordinate showed a statistically significant train–validation difference under a Holm-corrected Kolmogorov–Smirnov test. Together, these tests provide no evidence of a train–validation shift in the latent representation, supporting their use as a pooled sample for the unsupervised analyses.

A.2 Training-set and latent-capacity scans

The training-set scan uses 30 nested, species-balanced samples drawn from a fixed pool of 8,227 events per species, ranging from 100 to 22,212 total training images, with three seeds at each size. The largest changes occur in the low-data regime: weighted validation reconstruction improves throughout the scan, while unsupervised GMM agreement and proxy AUCs stabilise after the first few thousand images. From 12,342 to 22,212 images, the unsupervised species structure and five of six proxy AUCs show no significant trend; kaon topology increases from 0.896 to 0.907. The separately trained 9,419-image paper model lies within this stable region for all comparable clustering and proxy metrics.

The latent-capacity scan holds the training sample fixed at 12,342 images and varies the latent dimension from 4 to 128, again using three seeds at each value. Weighted validation reconstruction improves from 2880 at four dimensions to 1312 at 128 dimensions, reaching 95% of this gain near 48 dimensions. In contrast, GMM agreement shows no significant dependence on latent capacity, while increasing the latent size from 8 to 128 dimensions improves the six proxy AUCs by only 0.005–0.044. The KL term contributes only 0.5–6.5% of the loss function across the scan, consistent with weak regularisation.

We therefore use eight latent dimensions as a compact representation that retains the relevant particle and physical structure, rather than as the dimension that minimises reconstruction loss. Together, the scans show that the main result is stable to substantial changes in both training-set size and latent capacity.

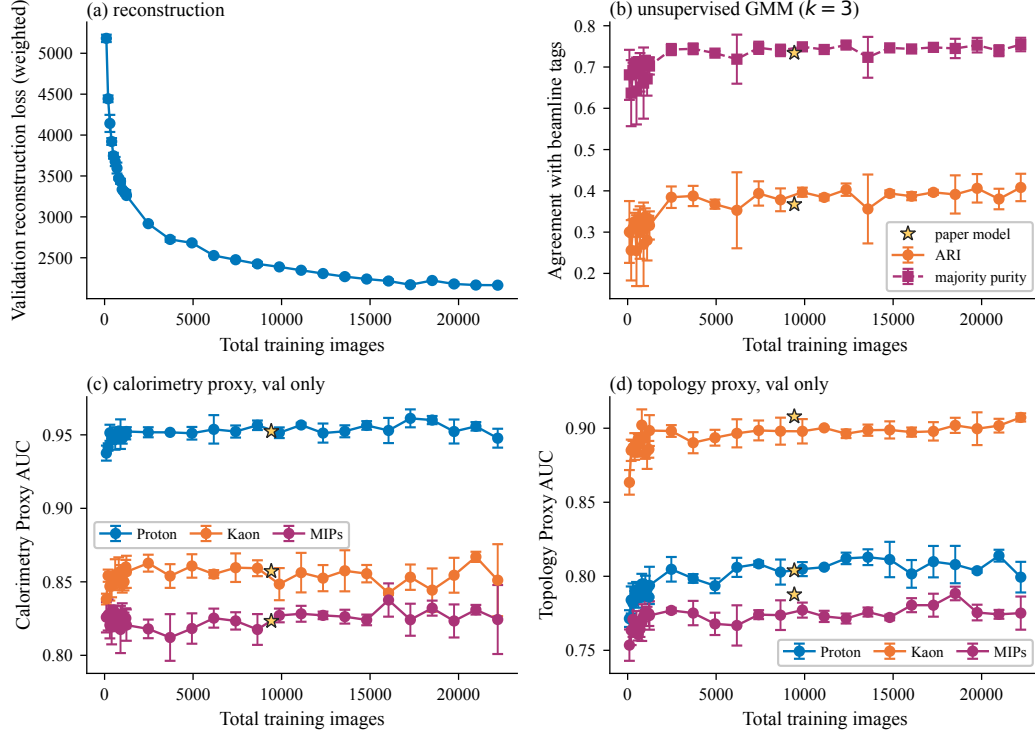


Figure 5: Training-set scan from 100 to 22,212 total training images, where the total is the sum of proton-, kaon-, and MIP-tagged events. Points and error bars show three-seed means and standard deviations. Stars mark the separately trained 9,419-image paper model on comparable clustering and proxy metrics. Proxy decodability and species structure stabilise while reconstruction continues to improve with more data.

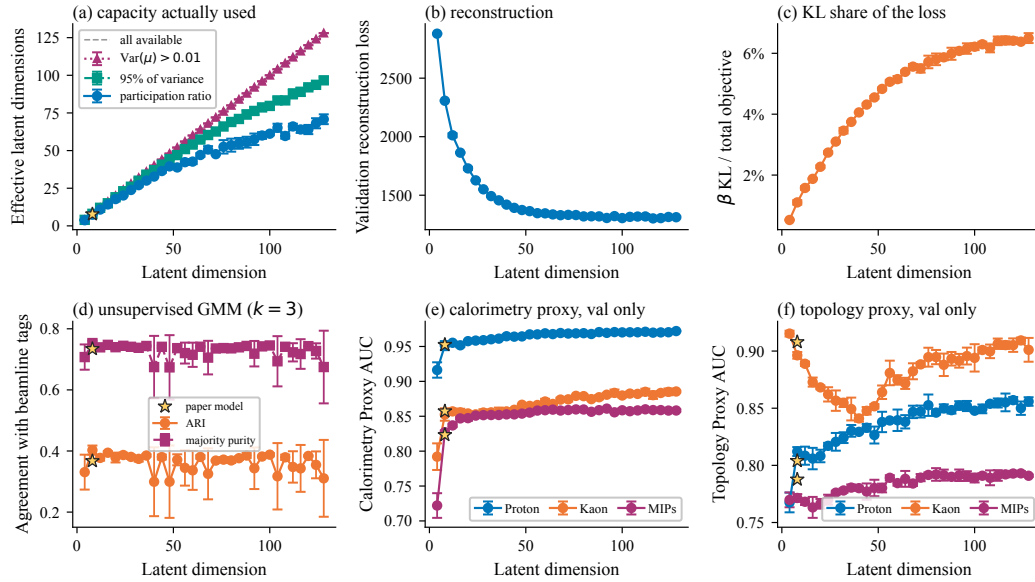


Figure 6: Latent-capacity scan using 12,342 training images. Points and error bars show three-seed means and standard deviations. Stars mark the separately trained eight-dimensional paper model on comparable representation, clustering, and proxy metrics. Reconstruction continues to improve with capacity while the recovered physical and species structure remains stable.

B Cluster-count scan and fit stability

Table 1 summarises the cluster-count scan. Overall purity is the fraction of events whose cluster majority agrees with the beamline tag. The fraction in 85%-pure clusters measures how much of the sample is assigned to a relatively clean population. It remains near one half after the dominant populations are resolved. Increasing k primarily subdivides existing populations.

Table 1: Selected points from the full-covariance GMM scan.

k	overall purity	best kaon purity	fraction in $\geq 85\%$	smallest cluster
3	0.734	0.635	0.000	8,691
8	0.729	0.873	0.495	2,088
12	0.737	0.965	0.464	1,124
15	0.803	0.945	0.457	852
37	0.818	0.985	0.554	196
50	0.821	1.000	0.555	63

For $k = 36$ to 41, five initialisation seeds give a mean mass shift of about +103 MeV, with a typical within- k standard deviation of 10.1 MeV. The mean pairwise adjusted Rand index between repeated partitions is only 0.62 to 0.66. Consequently, the aggregate shift is reproducible but individual fine clusters are not assigned physical status.

C Beamline-window context and comparison methods

The kaon tag is assigned by selecting a mass window from 350 to 650 MeV that lies in the valley between the light-particle and proton peaks (Fig. 7, left). A three-component Poisson fit to 4,009 high-quality beamline events finds a kaon peak at 488.3 ± 1.2 MeV with width 36.2 MeV and $\chi^2/\text{dof} = 0.91$. Within the selection window, the fitted fractions are 61.2% kaon, 25.1% proton, and 13.8% light-particle contributions. The 68% bootstrap interval on the proton fraction is 19.8% to 30.0%.

This fit precedes the exact stopping-track selection, so 25.1% is used as an approximate contamination scale rather than as the measured purity of the 8,227-event analysis sample. A separate latent-density check conditioned on the beamline tags gives calibrated proton fractions of 16.9% for the high-quality subset and 17.2% for the remaining events. This near equality supports transfer across the beamline quality flag, although the absolute latent estimate depends on the density model. Neither result enters the main unsupervised GMM or its mass validation.

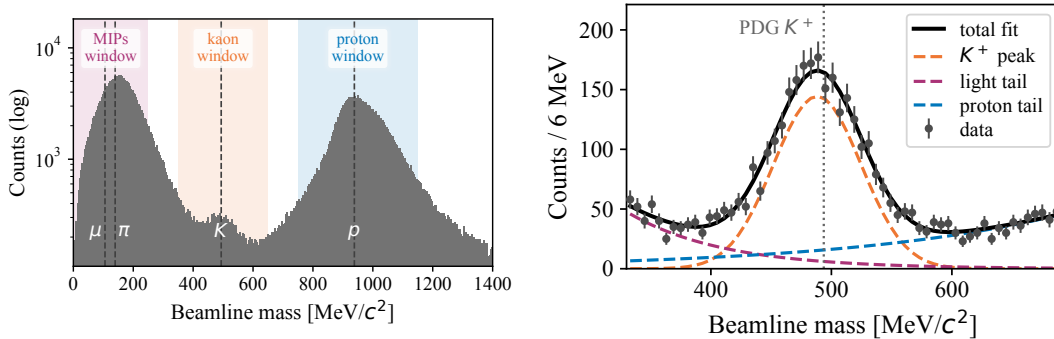


Figure 7: Beamline context. Left: the full non-negative mass spectrum and selection windows. Right: the high-quality three-component fit around the kaon window.

As a comparison, an anchored mixture uses the high-purity proton- and MIP-tagged selections to constrain two latent densities while refitting a free kaon cluster. It assigns 1,692 of 8,227 kaon-tagged events to the proton cluster and gives a +103.4 MeV mass shift, consistent with the fully unsupervised result. Because this method uses beamline tags during fitting, it is treated only as a supporting comparison.

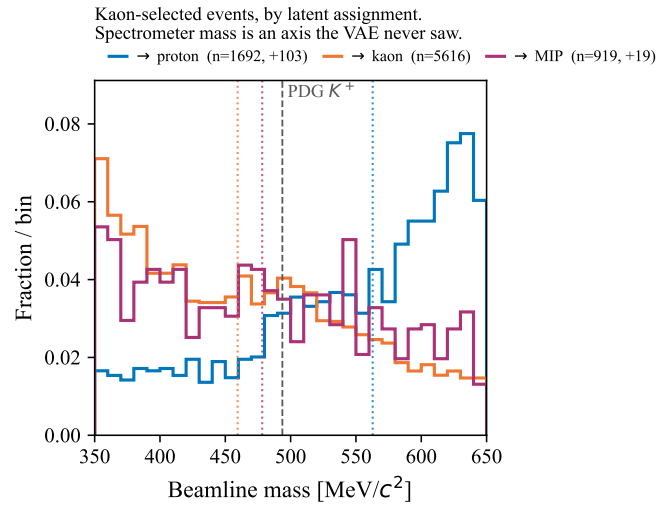


Figure 8: Beamline mass of kaon-tagged events after the anchored comparison fit. Mass is not used in the fit.